

In this work, we proposed a model which is **Iterative Error Feedback CNN-Transformer model with token pruning**. The goal is to capture both **local ECG morphology** and **global temporal dependencies**. We also introduced **token pruning** to reduce unnecessary computations. And an **iterative feedback mechanism** to improve predictions within the same training cycle.

Now let's break down this architecture step by step, starting with the CNN-Transformer hybrid.

So, The model starts with **1D CNN layers** to extract local spatial features like QRS, P, and T waves. These layers perform **hierarchical feature extraction**, from simple to complex patterns. Then features are passed to a **Transformer encoder**. Transformer captures **long-range dependencies using self-attention**. This combination allows us to learn both **local and global patterns effectively**.

To further enhance feature representation, we incorporate frequency-domain information along with time series information...

We used an **FFT branch** also to extract frequency-domain features. FFT converts ECG signals from **time domain to frequency domain**. This helps capture **rhythm periodicity and spectral characteristics**. Features from Time-series and FFT are **combined (concatenated)**. Thus, this allows model to learn both temporal and spectral characteristics.

After obtaining these rich features, we convert them into tokens for Transformer processing...

After all this, these features are converted into tokens using a **dense projection layer**. A **token** is a small piece of the ECG signal that the Transformer processes.

An ECG signal has **187 samples**. Instead of processing the whole signal together, the Transformer processes it as **small feature vectors called tokens**. A **positional encoding** is added to maintain sequence order.

Then we apply **token pruning**, where each token gets an importance score using Attention Score Guided Pruning.

Instead of assuming every token is equally important, the model calculates an **attention score** for every token. A lightweight **single-head proxy attention layer** compares each token with a **learnable global query vector**. The result is an importance score (attention score) for every token.

A **lightweight single-head proxy attention layer** computes an importance score for every token before pruning. A **learnable global query vector** is compared with every token, and the resulting scores are normalized using **Softmax** to obtain the attention score (α) for each token. The Top-60% tokens with the highest α values are retained.

Less important tokens are **down-weighted (soft pruning)**

This reduces **noise and computational cost**, focusing on important ECG segments

Finally, we have— iterative error feedback mechanism...

Each client trains on its own data and every client has a different data distribution (Non-IID). As a result, their learned weights move in **different directions**. When the server averages these weights, the global model is **not optimal for any client**. This is called **client drift**, one of the main problems in federated learning.

The model first makes a prediction. Then it looks at its own mistake, learns from that mistake immediately, corrects its internal features, and predicts again. The model first generates an **initial prediction** then computes the **prediction error (difference from true label)**

This error is projected into feature space and **fed back into the model** to refine features and reduce errors. A second Transformer pass performs **self-correction within the same iteration**.
This makes the model more accurate, faster to converge, and better at handling complex ECG patterns.